

The Economics of Brexit

Lecture 5: Regional disparities

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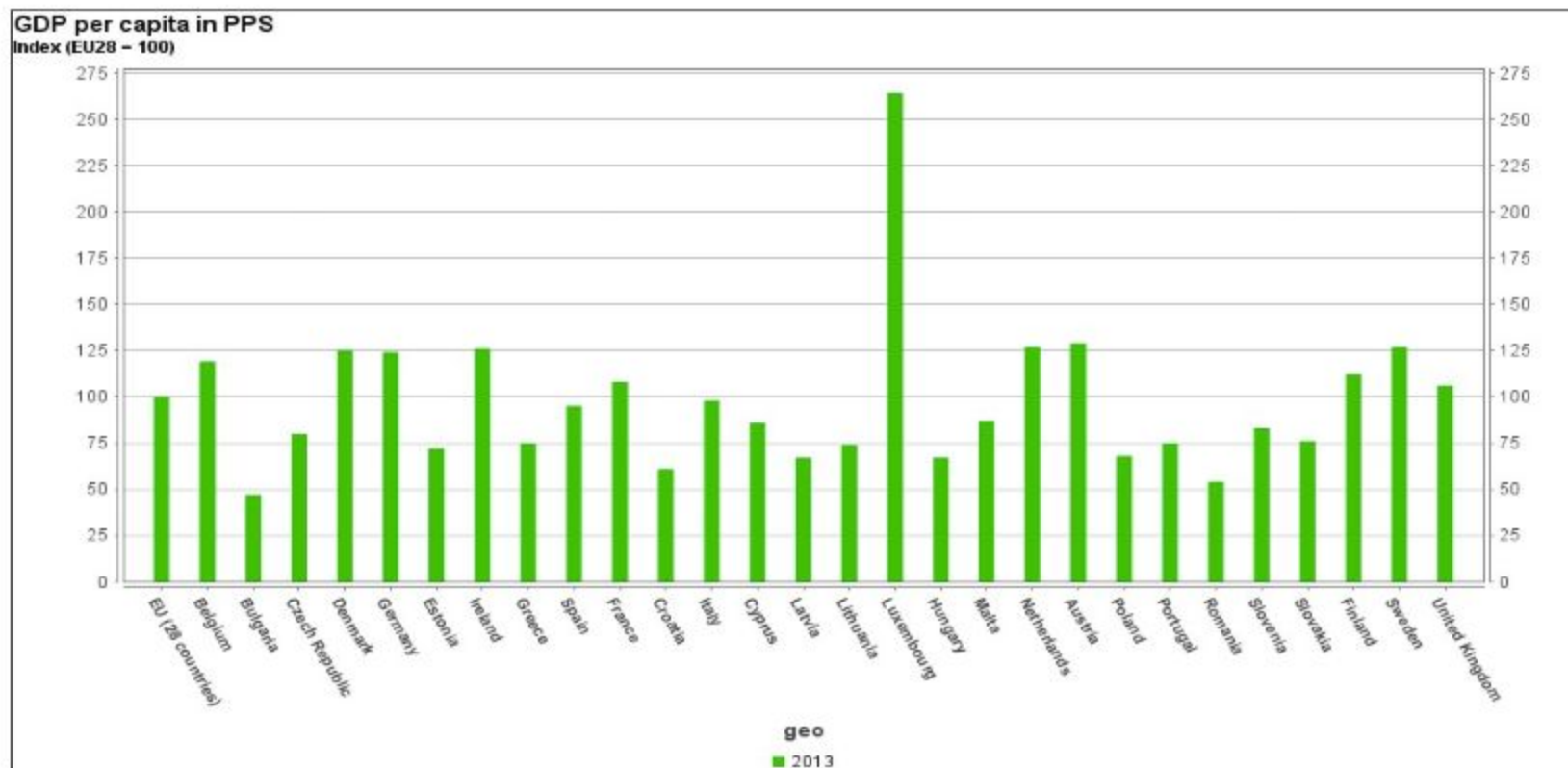
Macroeconomic Applications and Policy

Learning objectives

After this topic, you will be able to:

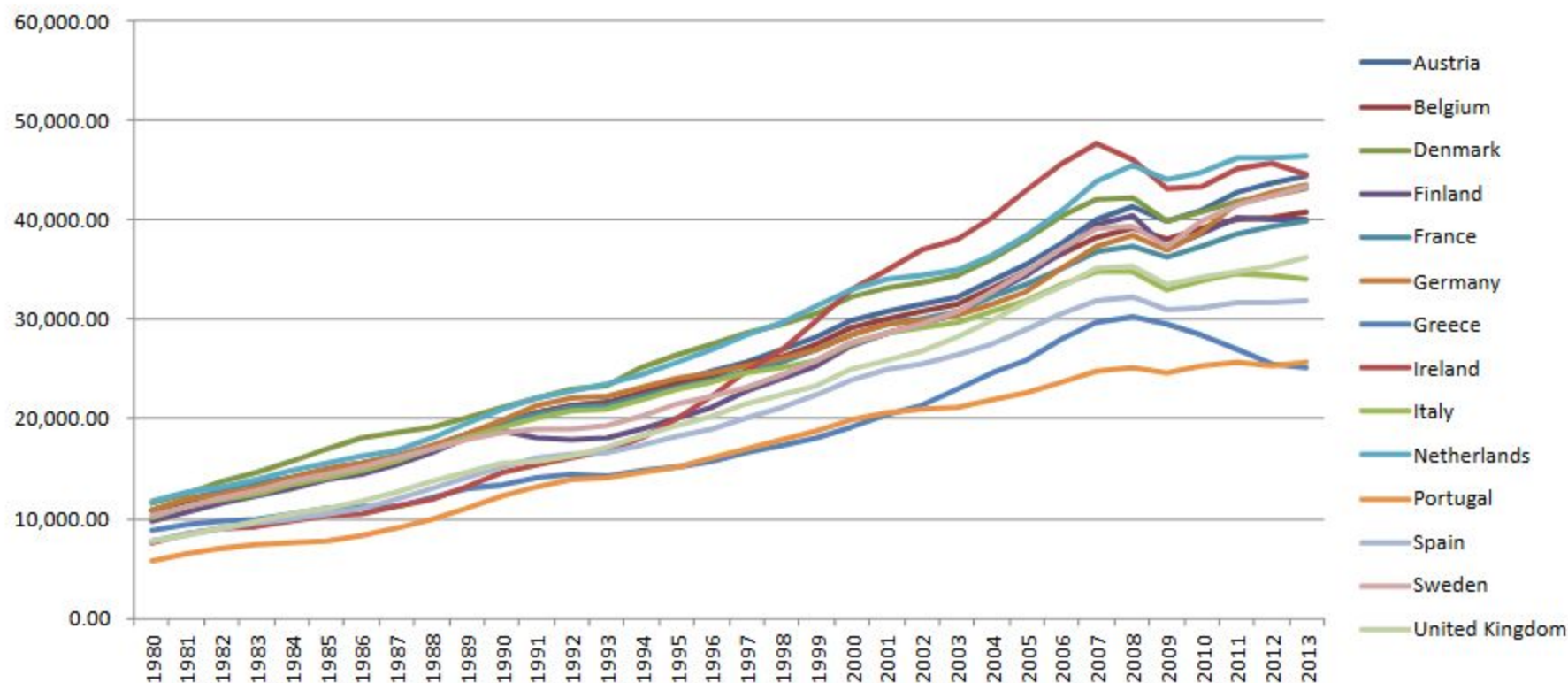
- Discuss evidence on differences between European countries and regions.
- Explain why these differences exist, using the Solow model and New Economic Geography.
- Explain how economic integration affects regional disparities.
- Discuss the aims of EU regional policy.

GDP per capita differences in the EU



Source: Eurostat

GDP per capita: EU15



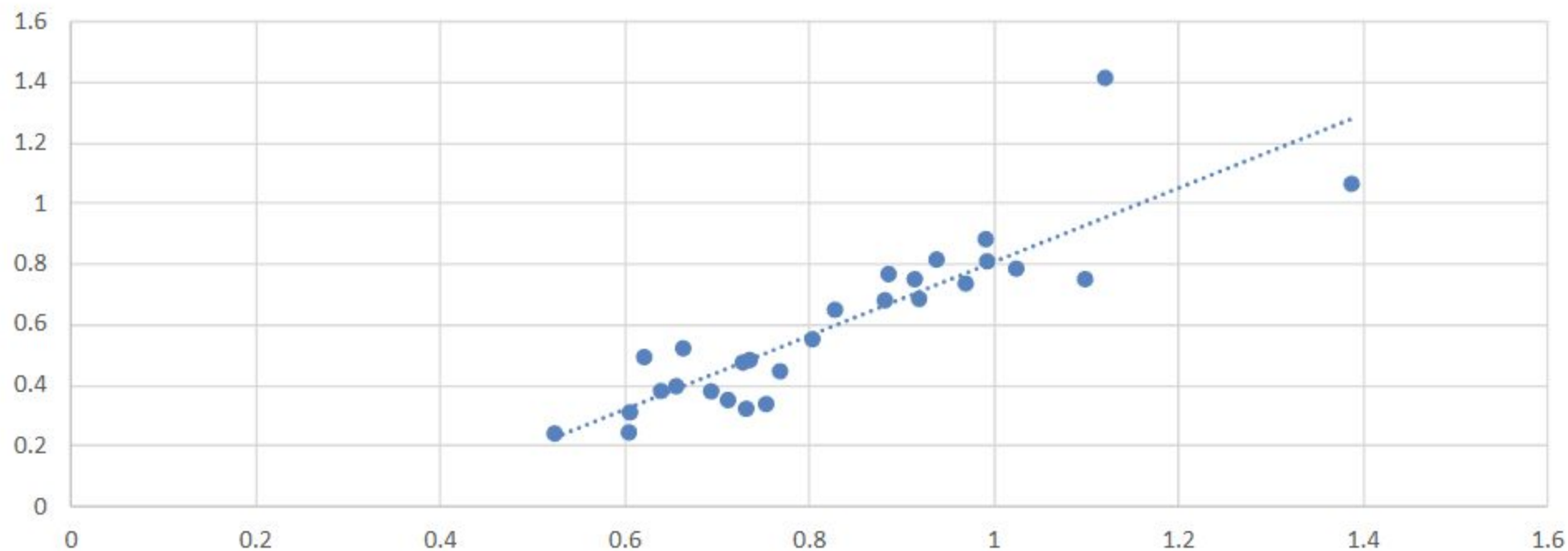
Source: IMF. Notes: Gross domestic product based on purchasing-power-parity (PPP) per capita GDP. Units: current international dollar.

Lack of cross-country convergence

- How can we reconcile this evidence with the Solow model?
- EU12 have not reached the steady state yet.
- Differences in the fundamentals.
- Productivity differences.

The role of productivity

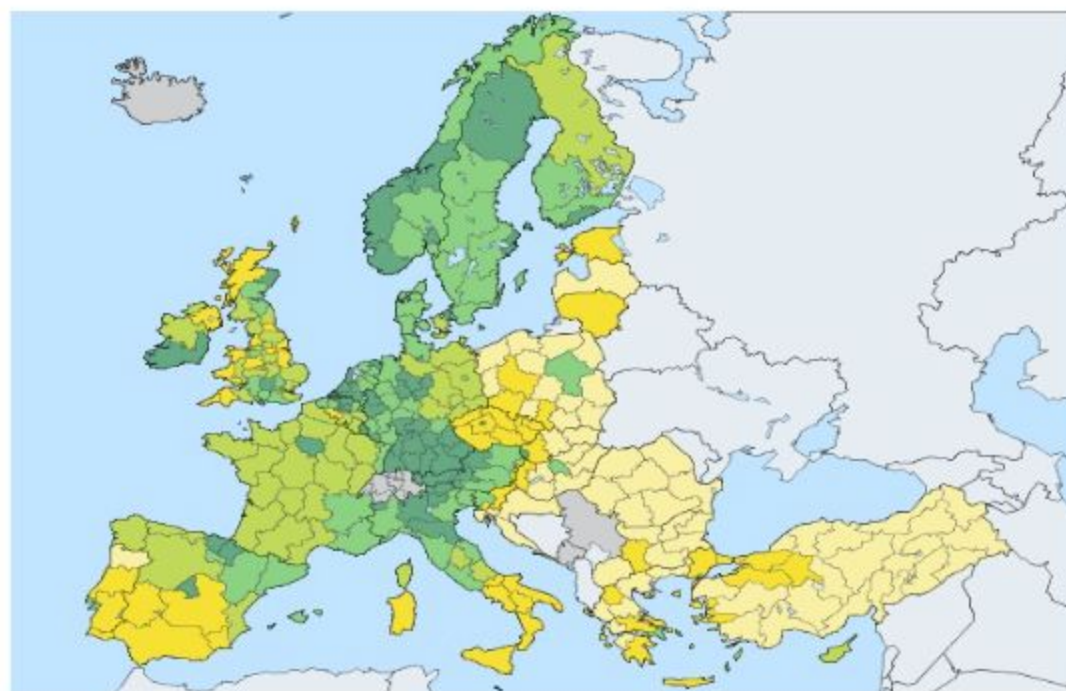
Real GDP per capita (y-axis) versus productivity (x-axis) in the EU 28, 2006



Source: Penn World Tables 8.1.

Notes: GDP per capita is output-side real GDP at current prices divided by the population. Total factor productivity and GDP per capita are relative to the US.

Regional disparities: GDP per capita

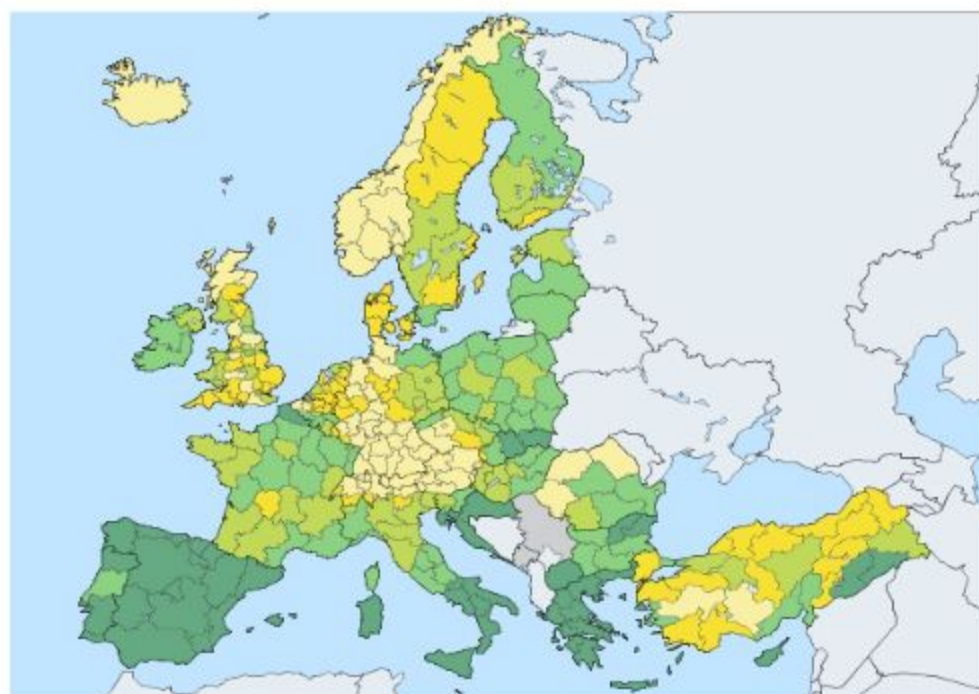


Regional GDP in purchasing-power standards per inhabitant, as a percentage of the EU28 average.

NUTS2 regions, 2011.

Source: Eurostat.

Regional disparities: unemployment



Legend

2.6 - 5.4

5.4 - 7.2

7.2 - 9.5

9.5 - 14.4

14.4 - 35.2

Not available

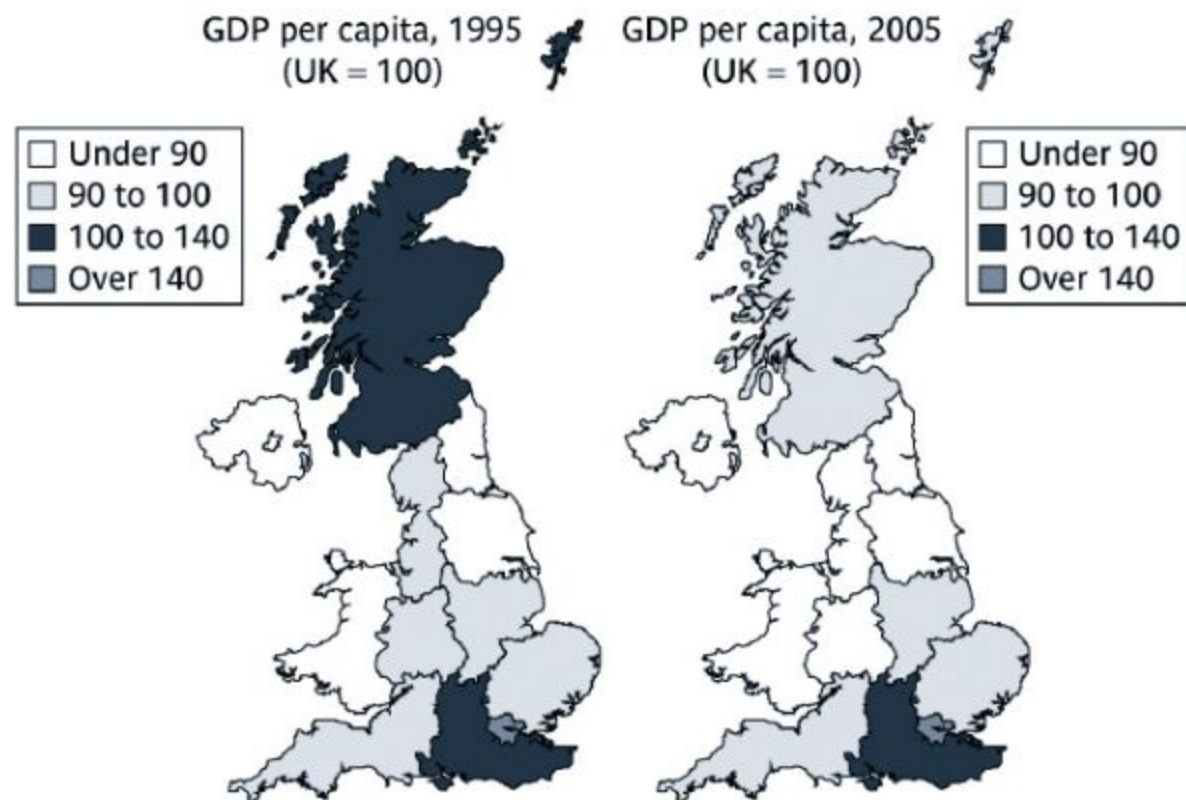
Minimum value: 2.6 Maximum value: 35.2

Regional unemployment rates.

NUTS2 regions, 2013.

Source: Eurostat.

Regional disparities over time



Regional disparities are increasing in the UK.

Source: Baldwin and Wyplosz (2012).

Regional disparities: theory

Main questions

- Which factors determine the location of economic activities within a country?
- Does European integration affect the location of economic activities in any way?
- “New Economic Geography”

Where to locate?

Dispersion forces

- Land prices.
- Congestion.
- Local competition.

Agglomeration forces

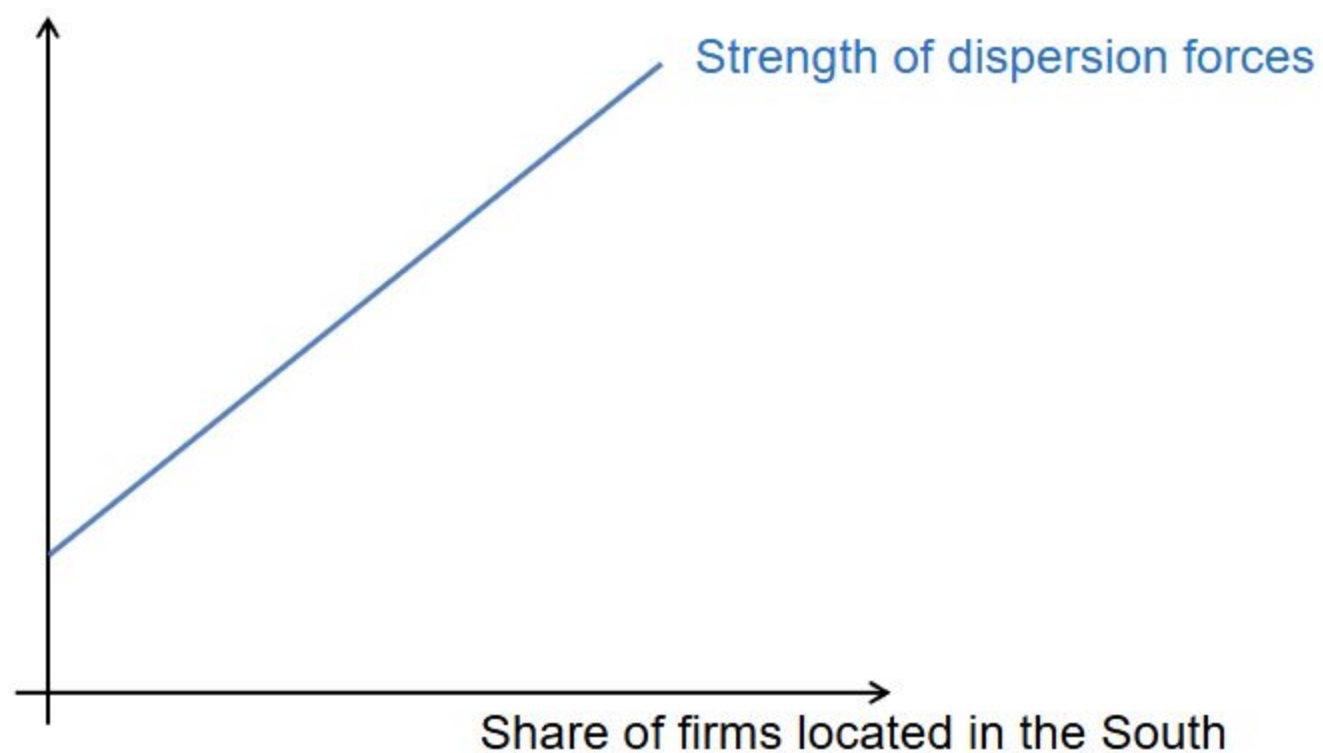
- Demand-linked.
- Cost-linked.

Two locations

NORTH

SOUTH

Dispersion forces



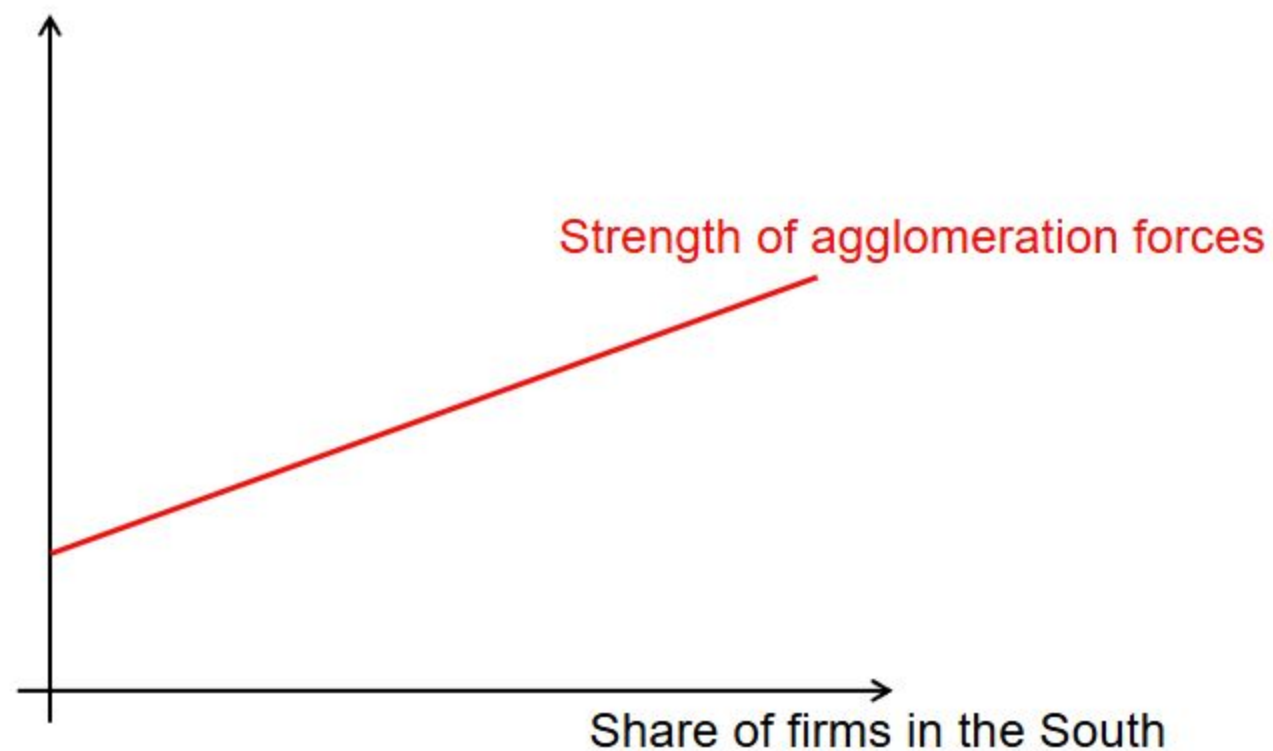
Demand-linked agglomeration forces

- Suppose the South's **share of the total market** suddenly increases.
- With shipping costs, firms relocate to the South.
- The share of firms in the South rises.
- Jobs, and therefore workers, move south.
- The South's market becomes even larger.

Supply-linked agglomeration forces

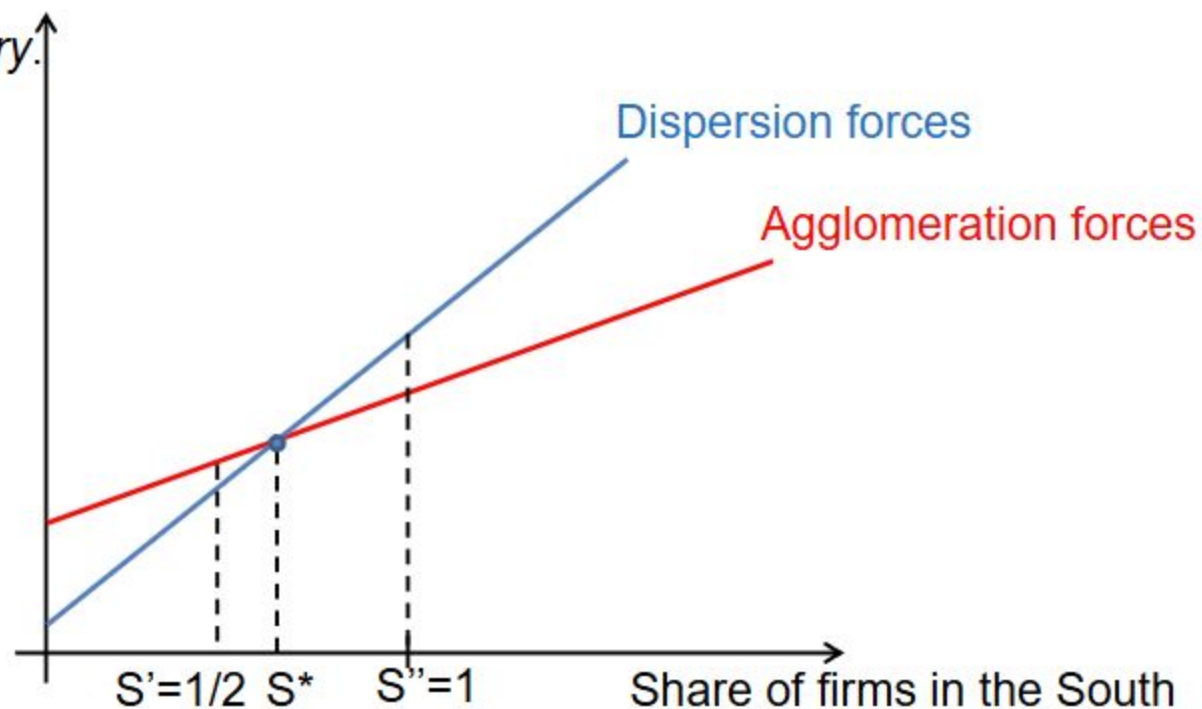
- Suppose the **cost of production** in the South suddenly falls.
- Firms relocate to the South.
- The share of firms in the South rises.
- More intermediate goods become available at a lower cost.
- Production costs in the South fall further.

Agglomeration forces



Firms' location in equilibrium

South is the *centre*.
North is the *periphery*.



European integration

European integration can be modelled as a reduction in shipping costs, increasing **market integration**.

Two opposing effects

- **Dispersion forces weaken:** the advantage of locating in the North falls.
- **Agglomeration forces weaken:** the advantage of locating in the South falls.

Dispersion force



Dispersion force: zero shipping costs

North

Firm 1

Firm 2

South

Firm 3

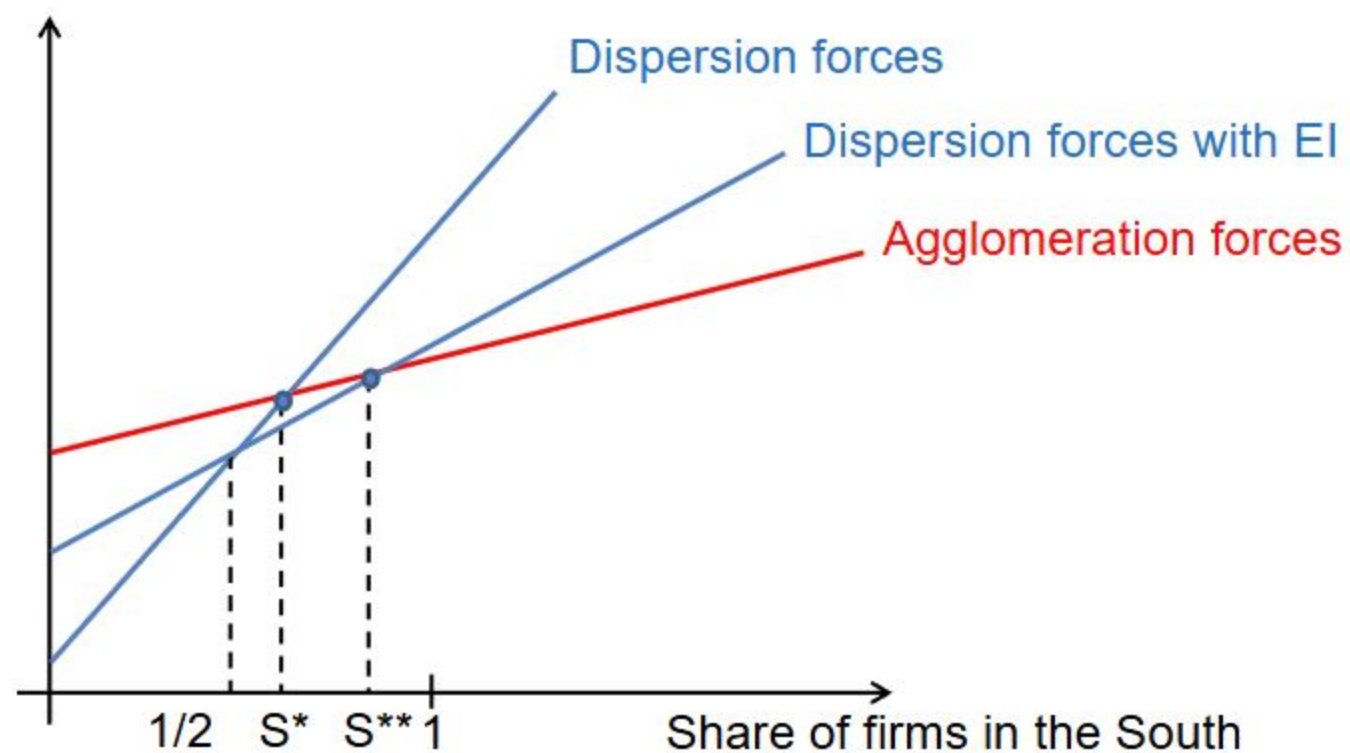
Firm 4

Firm 5

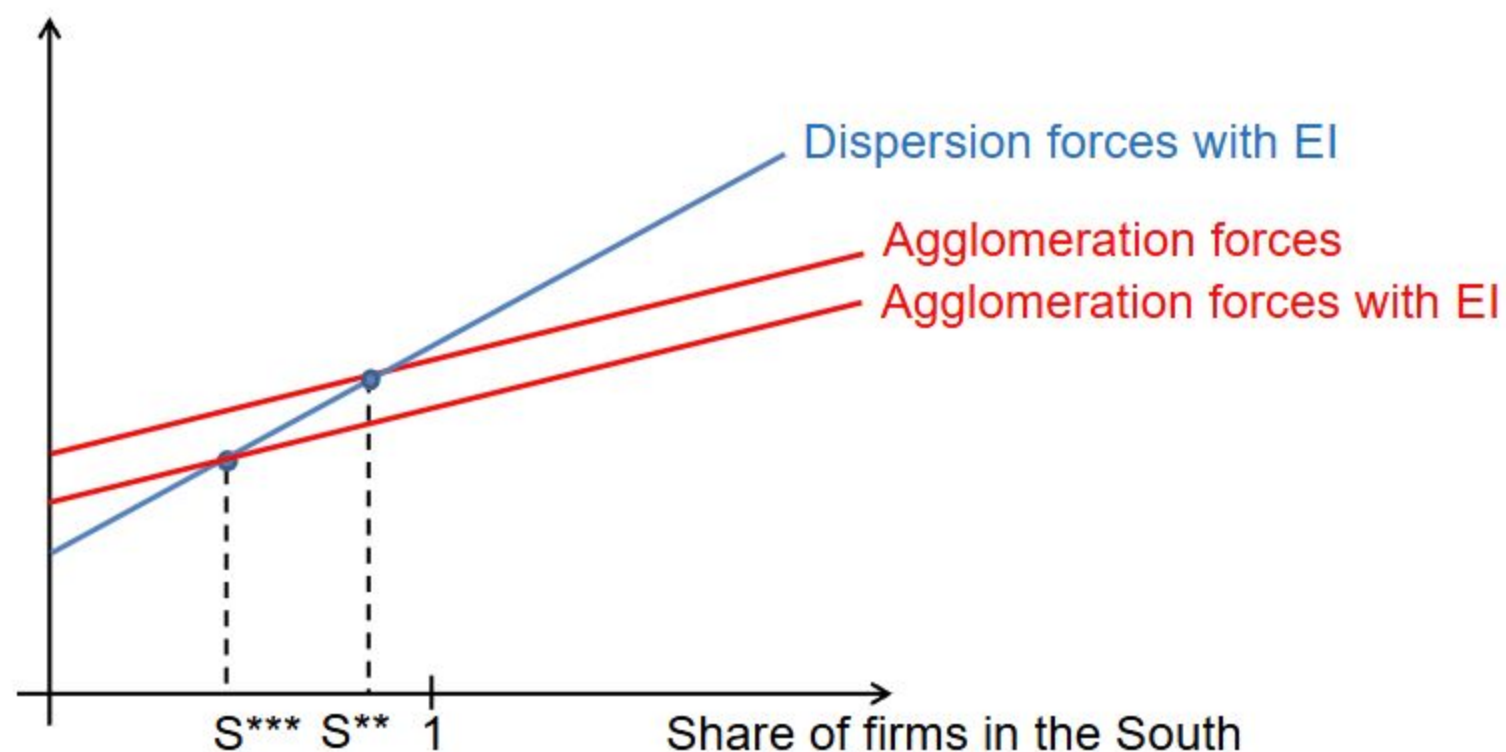
Firm 6

Shipping costs = 0

Dispersion force is weaker



Agglomeration force is weaker



Equilibrium effect?

- Since regional inequalities are increasing in Europe, then the incentives to locate in the periphery must have fallen by more than the incentives to locate in the core regions.

Regional policy

Non-interventionist view

- Regional policy penalises prosperous and productive regions.
- Disparities are a market outcome and must be tolerated.

Interventionist view

- Disparities result from market failures.
- These can be better addressed at the European level.
- As integration favours the core, lagging regions need support.

Regional policy: 2014–2020

One aim of the EU is to reduce regional disparities (EU Treaty).

Under the **2014–2020 strategy**, a third of the EU budget was committed to cohesion policy.

Three programmes

- Two-thirds to less-developed regions: below 75% of the EU27 average.
- 13% to transition regions: between 75% and 90% of the EU average.
- 20% to more-developed regions: above 90%.

The future of the EU

- Should the EU pursue the agenda of completing the political union or not?
- Is a federalist project for European countries a good idea to start with?

What is the optimal size of a country?

Contrasting views

“experience has shown that it is difficult, if not impossible, for a populous state to be run by good laws” (Aristotle, *The Politics*).

In the context of the US federation, Madison argued that a larger territory reduces the chance that “a majority of the whole will have a common motive to invade the rights of other citizens; or if such a common motive exists, it will be more difficult...to act in unison with each other” (Alesina et al., 2003, p. 5).

Alesina et al. (2005)

The optimal size of a country balances the benefits and costs of size.

Benefits

- Economies of scale in providing public goods.
- The market-size effect.

Cost

- Heterogeneity of preferences.

Economic integration changes this trade-off by reducing the benefits of size.

References

Essential reading

- Baldwin R. and Wyplosz C. (2022), Chapter 10.

Further reading

- [European Commission: Regional Policy](#)
- Alesina A., Spolaore E., Wacziarg R. (2005), Trade, growth and the size of countries, Handbook of Economic Growth, Vol. 1B, 1501-1514.